Marketing Mix Model

Data-Driven Budget Optimization

Omar Oudrari
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Marketing Analytics Project

MMM

SIMULATED DATA | FOR LEARNING USE ONLY

THE PROBLEM

We Spend $4.5M/Year Across 5 Channels But Don't Know What's Working

No Attribution

Cannot determine which channels actually drive revenue vs. merely correlate with it.

Budget Waste

TV gets 44% of spend but delivers the lowest ROI at 0.55x. Estimated misallocation costs $635K+/year.

Missed Revenue

Under-investment in high-ROI channels (Email at 7.48x, Display at 2.34x) leaves money on the table.

Bottom Line: Without a model, budget decisions rely on intuition, not evidence.

METHODOLOGY

A Rigorous 5-Step Modeling Pipeline

01

Raw Spend Data

Weekly channel investment data

02

Adstock Transform

Geometric decay for carryover

03

Saturation (Hill Fn)

Diminishing returns curve fitting

04

OLS Regression

With seasonality, holiday, competitor

05

Decompose & Optimize

Attribution + budget allocation

Key Assumptions

Adstock captures carryover effects (e.g., TV ad seen this week drives purchase next week). Saturation via Hill function models the point of diminishing returns per dollar spent. Together they remove bias that would distort a naive regression. Controls for seasonality, holidays, and competitor promotions isolate the true marketing effect.

DATA

2 Years of Weekly Data Across 5 Paid Channels

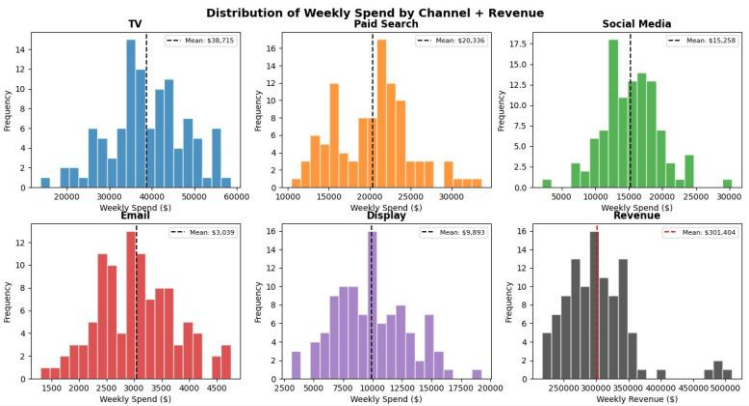
104
Weeks of data (2 yrs)

5
Marketing channels

3
Control variables

Channel	Avg Weekly Spend	Spend Range	% of Budget
TV	$38,715	$14K – $59K	44%
Paid Search	$20,336	$10K – $34K	23%
Social Media	$15,258	$2K – $30K	17%
Display	$9,893	$3K – $19K	11%
Email / CRM	$3,039	$1K – $5K	3%

Controls: seasonality index, holiday flags (Black Friday +75%, Christmas +60%), competitor promo (−8%)

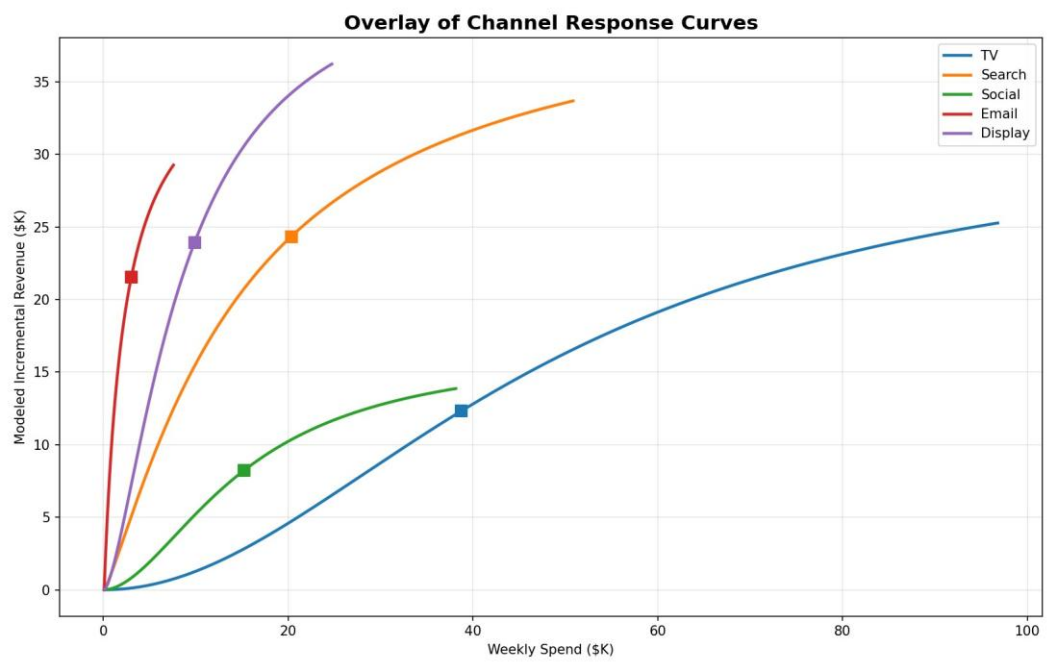


Mean Weekly Revenue: $301,404

Total Weekly Budget: $87,240

FINDINGS

Every Channel Shows Diminishing Returns — But at Different Rates



Email

Highest ROI at 7.48x — saturates fast due to small list size, but every dollar works hard.

Display

Strong 2.34x ROI — currently underfunded at 11% of budget. Room to scale.

Search

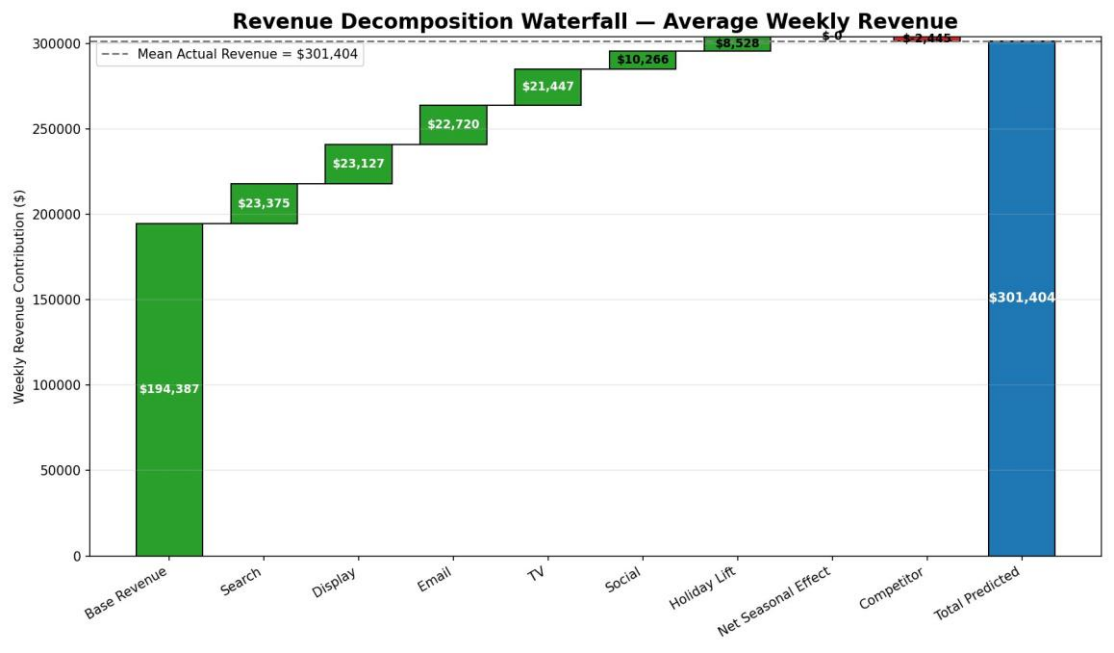
Moderate 1.15x ROI — solid performance channel with concave returns.

TV

Lowest ROI at 0.55x — gets 44% of budget but is deep in diminishing returns.

FINDINGS

Revenue Decomposition: What's Actually Driving Sales?



Contribution Breakdown

64% Base revenue (organic / brand)

8% Paid Search — top paid driver

8% Display — strong efficiency

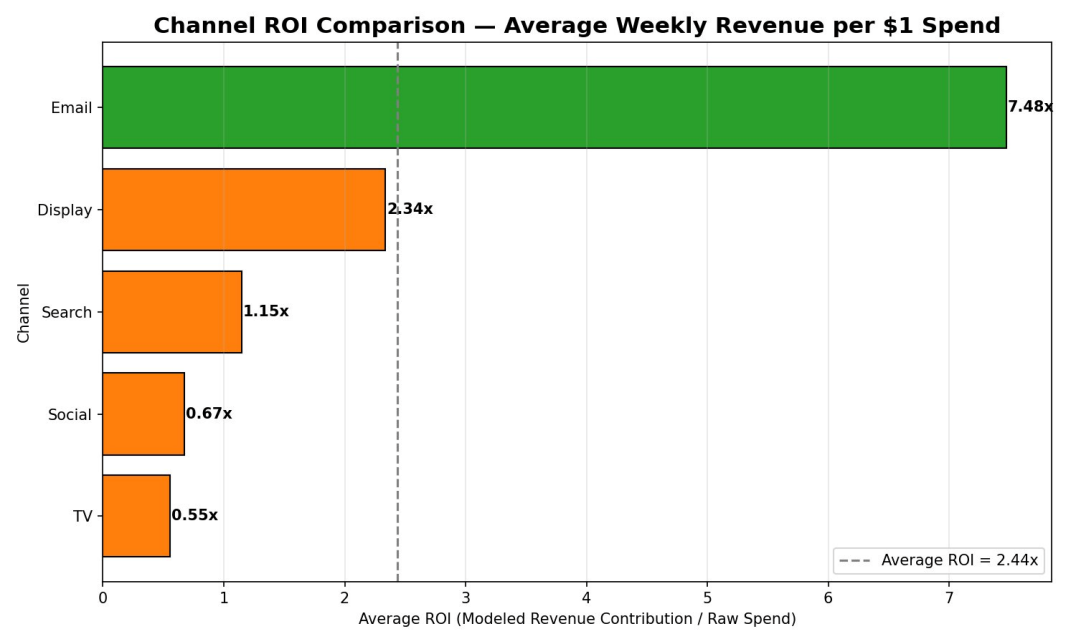
8% Email — highest per-dollar impact

7% TV — large spend, modest lift

3% Social + Holiday + Other

FINDINGS

Not All Channels Are Equal: ROI Varies 14x Across Channels



MOST EFFICIENT

Email / CRM

$7.48 return per $1 spent

Smallest budget, highest leverage. Constrained by list size, not spend ceiling.

LEAST EFFICIENT

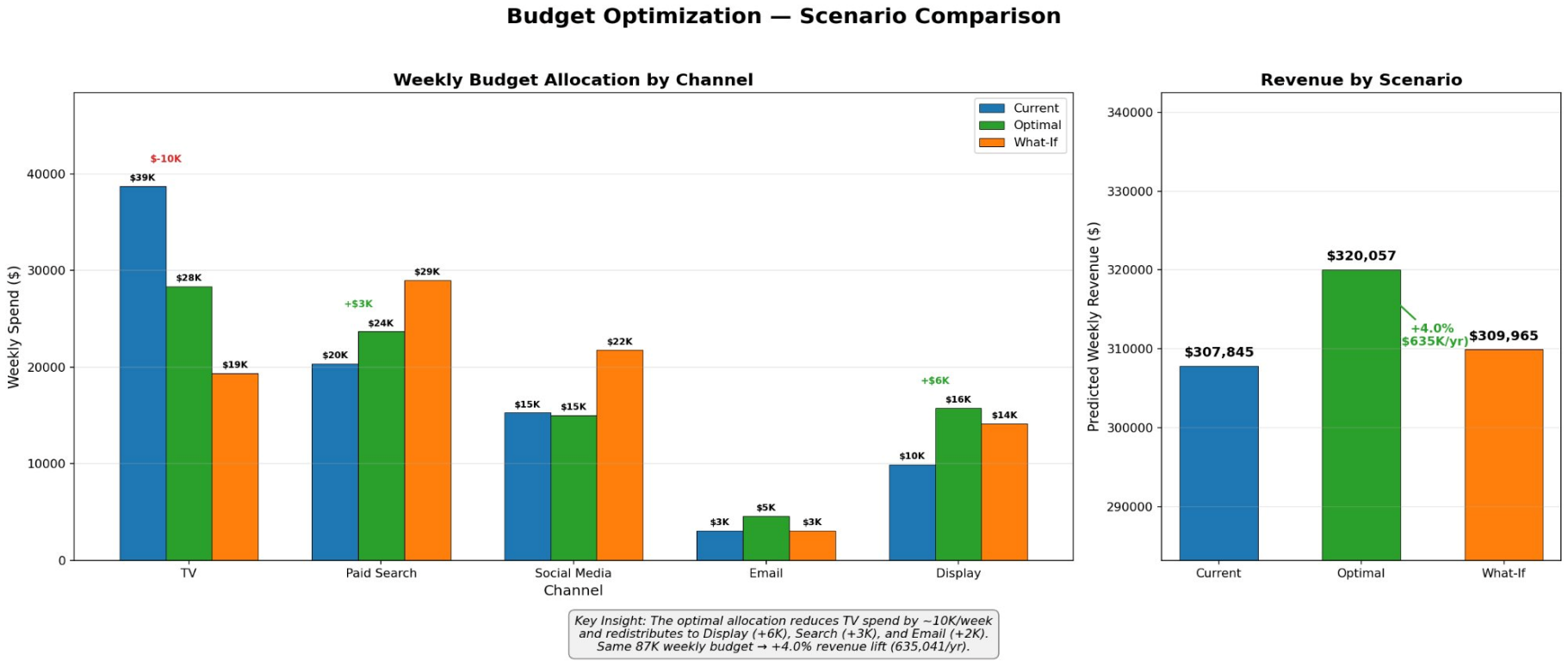
TV

$0.55 return per $1 spent

Commands 44% of budget but delivers the lowest marginal return of all channels.

OPTIMIZATION

Reallocation Could Generate +4% More Revenue With Same Budget



RECOMMENDATIONS

Reallocating Budget Could Unlock $635K in Incremental Annual Revenue

01 Cut TV from $39K to $28K/week

–$10.4K/wk redeployed

TV consumes 44% of budget at a 0.55x ROI. The model shows it is deep into diminishing returns. Redirect to higher-ROI channels.

02 Scale Display from $10K to $16K/week

+$5.8K/wk invested

Second-highest ROI at 2.34x and still in the steep part of its response curve. Room to scale before hitting saturation.

03 Increase Paid Search to $24K/week

+$3.3K/wk invested

Moderate 1.15x ROI with solid performance. Response curve shows it can absorb incremental spend efficiently up to ~$28K.

04 Grow Email from $3K to $5K/week

+$1.5K/wk invested

Highest marginal ROI at 7.48x. Constraint is list quality, not budget. Invest in CRM hygiene and list growth to unlock ceiling.

Next Steps

From Model to Impact: a 90-Day Roadmap

Days 1–30

Validate with Real Data

- Integrate actual revenue and spend feeds
- Re-run model on production data
- Audit adstock / saturation parameters with media team

Days 31–60

A/B Test Budget Shift

- Pilot Display + Email increase in 2 test markets
- Reduce TV spend in test cohort by $10K/wk
- Measure lift against holdout group

Days 61–90

Build Monitoring Dashboard

- Weekly ROI tracker per channel
- Alert when spend crosses saturation threshold
- Monthly model refresh pipeline

Questions? | Omar Oudrari | oudraroa@dukes.jmu.edu